EML4 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EML4 gene fusions from the msk_impact_50k_2026 study, covering 223 fusion events. 196/201 fusions with known frame status (97.5%) were in-frame. 119/223 fusions (53.4%) retained the HELP motif domain. Domain retention was tested with Fisher's exact test (p=0.999931, not statistically significant at alpha=0.05). No literature benchmark is configured for EML4.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 223 analyzed fusion events, identifying 3 distinct fusion partner genes. The most frequent partner was ALK, observed in 221 events.

Domain retention

HELP motif domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 95/196 (48.5%) of in-frame fusions retained the domain, p=0.999931 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Domain-disruption analysis was skipped: EML4 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

34% of 182 fusions with this gene as the 5' partner join at exon 13, against 4% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 184 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 223 aa (exon 6/7 boundary; label-permutation p=0.000999001). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 0 aa from the nearest configured domain boundary at 223 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=97) and "not_retained" (n=87). For Frame_status, retained: 78/82 (95.1%, 95% CI 88.1%-98.1%); not_retained: 84/84

(100.0%, 95% CI 95.6%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 67.6146 vs 62.3103, p=0.604098 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for EML4 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (41 row(s) in genomic_bin_recurrence, 2 row(s) in exact_position_recurrence, 7 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: EML4 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: EML4 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (96 row(s) in window_scan, 42 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ALK	221	0.9910313901345291	0.5000004342923104	None	0.6074259074259074	0.9282923589636631	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.7578158095296108	1
NTRK1	1	0.004484304932735426	0.5000004342923104	None	0.6909090909090909	0.9282923589636631	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.4035137494820352	2
RET	1	0.004484304932735426	0.5000004342923104	None	0.0	0.9282923589636631	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2739682949365807	3

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
70	0	97	15	72	0.0	6.59428167346614e-06	5.180832505991869
113	0	97	16	71	0.0	2.809398109405692e-06	5.551386714174274
223	1	96	86	1	0.0001211240310077519 4	7.679185037022372e-51	50.11468486756019
497	63	34	87	0	0.0	2.0286771032054753e-11	10.692787072461782
501	64	33	87	0	0.0	5.0276700238238306e-11	10.29863323367803
532	65	32	87	0	0.0	1.2376021720416103e-10	9.90741893710373
547	67	30	87	0	0.0	7.347225450204457e-10	9.133876633590017
656	68	29	87	0	0.0	8.528605440240532e-10	9.069121976935888
686	76	21	87	0	0.0	4.937220617042471e-07	6.306517466132007
718	81	16	87	0	0.0	2.161926453446647e-05	4.665159084390838
748	92	5	87	0	0.0	0.060941084468509925	1.21508982161812
752	93	4	87	0	0.0	0.1231291066313099	0.9096392715597116
759	94	3	87	0	0.0	0.2481290092658589	0.6053224585105558
771	95	2	87	0	0.0	0.4987526728439059	0.30211476389015995
778	96	1	87	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

78	19
84	3

frame_domain_contingency_tables_by_tumor_type

stratum	table
ADNOS	[0, 0], [1, 0]
CSCLC	[0, 1], [0, 0]
CUP	[0, 0], [1, 0]
IDC	[0, 0], [1, 0]
LUAD	[75, 17], [70, 2]
LUAS	[1, 0], [0, 0]
LUSC	[0, 0], [1, 0]
MACR	[0, 0], [0, 1]
MPNST	[0, 0], [1, 0]
NSCLC	[1, 0], [0, 0]
NSCLCPD	[0, 0], [1, 0]
OSACA	[0, 0], [1, 0]
PAAD	[0, 0], [2, 0]
PDC	[0, 0], [1, 0]
SARCL	[0, 0], [1, 0]
SDCA	[0, 0], [1, 0]
THHC	[0, 1], [0, 0]
THPD	[1, 0], [0, 0]
URCC	[0, 0], [2, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	HELP motif	184	97	70	17	0.01058681185722928

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
6	166	184	0.9021739130434783

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ALK	221	221	220	182
NTRK1	1	1	1	1
RET	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
2	42523000	42524000	14	EVT-P-0002217-T02-IM3-5, EVT-P-0003446-T01-IM5-14, EVT-P-0003913-T01-IM3-15, EVT-P-0008332-T01-IM5-26, EVT-P-0013864-T01-IM5-47, ... (14 total)
2	42525000	42526000	12	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0003383-T01-IM5-13, EVT-P-0010177-T01-IM5-35, EVT-P-0012679-T01-IM5-42, ... (12 total)
2	42524000	42525000	10	EVT-P-0006950-T01-IM5-21, EVT-P-0012873-T02-IM6-135, EVT-P-0014385-T01-IM6-51, EVT-P-0016580-T01-IM6-61, EVT-P-0019513-T01-IM6-72, ... (10 total)
2	42527000	42528000	10	EVT-P-0018969-T01-IM6-70, EVT-P-0019491-T01-IM6-71, EVT-P-0027119-T01-IM6-97, EVT-P-0028901-T01-IM6-106, EVT-P-0031368-T04-IM6-139, ... (10 total)
2	42553000	42554000	10	EVT-P-0005743-T02-IM5-152, EVT-P-0022138-T01-IM6-86, EVT-P-0025095-T01-IM6-147, EVT-P-0026657-T01-IM6-95, EVT-P-0027853-T01-IM6-99, ... (10 total)
2	42528000	42529000	9	EVT-P-0002613-T01-IM3-1, EVT-P-0005321-T01-IM5-18, EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, EVT-P-0009102-T01-IM5-27, ... (9 total)

chromosome	bin_start	bin_end	n_events	event_ids
2	42526000	42527000	8	EVT-P-0006370-T01-IM5-20, EVT-P-0021693-T01-IM6-84, EVT-P-0025040-T01-IM6-146, EVT-P-0032683-T01-IM6-118, EVT-P-0036106-T01-IM6-123, ... (8 total)
2	42494000	42495000	7	EVT-P-0000557-T01-IM3-125, EVT-P-0009353-T01-IM5-33, EVT-P-0013202-T01-IM5-45, EVT-P-0014270-T01-IM6-49, EVT-P-0021495-T01-IM6-83, ... (7 total)
2	42492000	42493000	6	EVT-P-0003206-T01-IM5-10, EVT-P-0003323-T01-IM5-12, EVT-P-0007211-T01-IM5-24, EVT-P-0015921-T03-IM6-56, EVT-P-0022926-T01-IM6-88, ... (6 total)
2	42496000	42497000	6	EVT-P-0023962-T01-IM6-90, EVT-P-0034334-T01-IM6-150, EVT-P-0037404-T01-IM6-157, EVT-P-0037925-T01-IM6-160, EVT-P-0051231-T01-IM6-193, ... (6 total)
2	42501000	42502000	6	EVT-P-0009301-T01-IM5-28, EVT-P-0009339-T01-IM5-30, EVT-P-0010041-T01-IM5-34, EVT-P-0011203-T01-IM5-39, EVT-P-0015961-T01-IM6-57, ... (6 total)
2	42506000	42507000	6	EVT-P-0009343-T01-IM5-32, EVT-P-0011018-T01-IM5-38, EVT-P-0015110-T01-IM6-52, EVT-P-0040574-T01-IM6-166, EVT-P-0044042-T01-IM6-127, ... (6 total)
2	42552000	42553000	6	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0015152-T02-IM6-53, EVT-P-0029174-T01-IM6-107, EVT-P-0049511-T01-IM6-192, ... (6 total)
2	42493000	42494000	5	EVT-P-0012778-T01-IM5-43, EVT-P-0013299-T02-IM6-143, EVT-P-0019972-T01-IM6-75, EVT-P-0040600-T01-IM6-168, EVT-P-0049375-T01-IM6-190
2	42499000	42500000	5	EVT-P-0016960-T01-IM6-62, EVT-P-0017978-T01-IM6-64, EVT-P-0026650-T02-IM6-94, EVT-P-0035085-T03-IM6-153, EVT-P-0069617-T01-IM7-220
2	42503000	42504000	5	EVT-P-0024136-T01-IM6-91, EVT-P-0035121-T01-IM6-122, EVT-P-0043350-T01-IM6-173, EVT-P-0045341-T01-IM6-181, EVT-P-0069396-T02-IM7-218

chromosome	bin_start	bin_end	n_events	event_ids
2	42504000	42505000	5	EVT-P-0003297-T01-IM5-11, EVT-P-0020138-T01-IM6-77, EVT-P-0045243-T01-IM6-180, EVT-P-0057391-T02-IM6-199, EVT-P-0064716-T02-IM7-210
2	42505000	42506000	5	EVT-P-0015197-T01-IM6-54, EVT-P-0028287-T01-IM6-100, EVT-P-0031406-T01-IM6-113, EVT-P-0063585-T01-IM7-208, EVT-P-0065210-T01-IM7-211
2	42507000	42508000	5	EVT-P-0029550-T01-IM6-109, EVT-P-0029971-T01-IM6-111, EVT-P-0040576-T01-IM6-167, EVT-P-0054263-T01-IM6-197, EVT-P-0059257-T02-IM7-201
2	42543000	42544000	4	EVT-P-0013453-T01-IM5-46, EVT-P-0019786-T01-IM6-74, EVT-P-0021257-T01-IM6-82, EVT-P-0044432-T01-IM6-222
2	42544000	42545000	4	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0022101-T01-IM6-85
2	42476000	42477000	3	EVT-P-0001766-T04-IM6-151, EVT-P-0064342-T01-IM7-209, EVT-P-0069535-T01-IM7-219
2	42498000	42499000	3	EVT-P-0000732-T01-IM3-65, EVT-P-0019533-T01-IM6-73, EVT-P-0023368-T01-IM6-89
2	42522000	42523000	3	EVT-P-0015434-T02-IM6-55, EVT-P-0036247-T01-IM6-124, EVT-P-0040553-T01-IM6-165
2	42473000	42474000	2	EVT-P-0002266-T01-IM3-6, EVT-P-0059139-T01-IM7-224
2	42474000	42475000	2	EVT-P-0028339-T01-IM6-101, EVT-P-0067093-T02-IM7-216
2	42477000	42478000	2	EVT-P-0020825-T03-IM6-81, EVT-P-0028541-T01-IM6-102
2	42479000	42480000	2	EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142
2	42482000	42483000	2	EVT-P-0024907-T01-IM6-92, EVT-P-0030590-T01-IM6-112
2	42497000	42498000	2	EVT-P-0000474-T01-IM3-9, EVT-P-0014310-T01-IM6-50
2	42500000	42501000	2	EVT-P-0018242-T01-IM6-67, EVT-P-0022398-T01-IM6-87
2	42502000	42503000	2	EVT-P-0005305-T01-IM5-17, EVT-P-0017592-T01-IM5-63
2	42546000	42547000	2	EVT-P-0010978-T01-IM5-36, EVT-P-0049318-T01-IM6-189

chromosome	bin_start	bin_end	n_events	event_ids
2	42550000	42551000	2	EVT-P-0013886-T01-IM5-225, EVT-P-0026407-T01-IM6-93
2	42475000	42476000	1	EVT-P-0031824-T01-IM6-141
2	42478000	42479000	1	EVT-P-0020584-T01-IM6-79
2	42486000	42487000	1	EVT-P-0043594-T01-IM6-175
2	42495000	42496000	1	EVT-P-0066568-T02-IM7-213
2	42529000	42530000	1	EVT-P-0034450-T01-IM6-120
2	42540000	42541000	1	EVT-P-0019560-T01-IM6-148
2	42545000	42546000	1	EVT-P-0045912-T02-IM6-182

exact_position_recurrence

chromosome	position	n_events	event_ids
2	42526326	2	EVT-P-0036106-T01-IM6-123, EVT-P-0062999-T02-IM7-207
2	42528243	2	EVT-P-0005321-T01-IM5-18, EVT-P-0037982-T01-IM6-161

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
223	71	71	15835	False
497	63	61	5557	False
70	15	15	8802	False
748	11	11	434	False
686	8	8	1180	False
718	5	5	4884	False
547	2	2	554	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	6	70	182	0.38461538461538464	0.09892217476822775	23.86704591410269
five_prime	13	62	182	0.34065934065934067	0.035907082763987526	41.95523705350065
five_prime	2	15	182	0.08241758241758242	0.06737786383494997	0.6083210757374271
five_prime	20	11	182	0.06043956043956044	0.004209584617552804	9.29402367495309
five_prime	18	8	182	0.04395604395604396	0.008976770690996881	3.5674380734674678
five_prime	19	5	182	0.027472527472527472	0.04926500775127299	0.023150364426656408

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	21	5	182	0.027472527472527472	0.016740298653790772	0.718359565842393
five_prime	14	4	182	0.02197802197802198	0.01141551112445542	0.8064526183485669
five_prime	3	1	182	0.005494505494505495	0.028308996991403133	0.002339281426407406
five_prime	17	1	182	0.005494505494505495	0.07033131330461584	7.473390368054308e-07
three_prime	7	1	2	0.5	0.09952266864380296	0.7232152933748686
three_prime	14	1	2	0.5	0.036005122580407965	1.1504953554674546

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-130	70	200	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
-87	113	200	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
-30	70	100	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
13	113	100	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
20	70	50	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
23	223	200	87	97	1	96	0.00012112403100775194	7.679185037022372e-51	50.11468486756019	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, ... (87 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
45	70	25	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
63	113	50	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
70	95	25	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
70	120	50	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
70	170	100	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
70	270	200	87	97	1	96	0.00012112403100775194	7.679185037022372e-51	50.11468486756019	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, ... (87 total)
113	313	200	72	112	1	96	0.002347417840375587	6.248379429212107e-34	33.20423260604981	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (72 total)
123	223	100	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
173	223	50	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
198	223	25	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
223	248	25	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
223	273	50	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
223	323	100	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
223	423	200	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
297	497	200	63	121	62	35	152.34285714285716	2.115676160934519e-22	21.67455080746973	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (63 total)
301	501	200	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)
332	532	200	65	119	64	33	166.78787878787878	1.7093128523025282e-23	22.76717844190544	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (65 total)
347	547	200	67	117	66	31	183.09677419354838	1.2835177750757548e-24	23.89159811252907	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (67 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
397	497	100	63	121	62	35	152.34285714285716	2.115676160934519e-22	21.67455080746973	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (63 total)
401	501	100	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)
432	532	100	65	119	64	33	166.78787878787878	1.7093128523025282e-23	22.76717844190544	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (65 total)
447	497	50	63	121	62	35	152.34285714285716	2.115676160934519e-22	21.67455080746973	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (63 total)
447	547	100	67	117	66	31	183.09677419354838	1.2835177750757548e-24	23.89159811252907	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (67 total)
451	501	50	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)
456	656	200	68	116	67	30	192.06666666666666	3.411263875176258e-25	24.467084684843417	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (68 total)
472	497	25	63	121	62	35	152.34285714285716	2.115676160934519e-22	21.67455080746973	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (63 total)
476	501	25	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
482	532	50	65	119	64	33	166.78787878787878	1.7093128523025282e-23	22.76717844190544	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (65 total)
486	686	200	76	108	75	22	293.1818181818182	4.406093942417991e-30	29.35594624748511	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002408-T02-IM5-8, EVT-P-0002613-T01-IM3-1, ... (76 total)
497	522	25	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)
497	547	50	67	117	66	31	183.09677419354838	1.2835177750757548e-24	23.89159811252907	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (67 total)
497	597	100	67	117	66	31	183.09677419354838	1.2835177750757548e-24	23.89159811252907	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (67 total)
497	697	200	76	108	75	22	293.1818181818182	4.406093942417991e-30	29.35594624748511	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002408-T02-IM5-8, EVT-P-0002613-T01-IM3-1, ... (76 total)
501	551	50	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, EVT-P-0034450-T01-IM6-120, EVT-P-0046720-T01-IM6-186
501	601	100	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, EVT-P-0034450-T01-IM6-120, EVT-P-0046720-T01-IM6-186
501	701	200	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, ... (13 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
518	718	200	17	167	17	80	None	1.0256347894034088e-05	4.98900725636106	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0007705-T01-IM5-25, EVT-P-0010978-T01-IM5-36, ... (17 total)
532	732	200	17	167	17	80	None	1.0256347894034088e-05	4.98900725636106	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0007705-T01-IM5-25, EVT-P-0010978-T01-IM5-36, ... (17 total)
547	747	200	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (16 total)
548	748	200	25	159	25	72	None	2.165834310081195e-08	7.664374770688462	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (25 total)
552	752	200	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)
559	759	200	27	157	27	70	None	4.36030226790699e-09	8.36048340322365	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (27 total)
571	771	200	28	156	28	69	None	1.9355912689204797e-09	8.71318634557778	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (28 total)
578	778	200	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)
581	781	200	30	154	30	67	None	7.347225450204457e-10	9.133876633590017	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
586	686	100	9	175	9	88	None	0.003556113151803106	2.4490244286855454	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019560-T01-IM6-148, ... (9 total)
618	718	100	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (14 total)
636	686	50	9	175	9	88	None	0.003556113151803106	2.4490244286855454	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019560-T01-IM6-148, ... (9 total)
648	748	100	25	159	25	72	None	2.165834310081195e-08	7.664374770688462	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (25 total)
652	752	100	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)
656	706	50	9	175	9	88	None	0.003556113151803106	2.4490244286855454	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019560-T01-IM6-148, ... (9 total)
656	756	100	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)
656	856	200	30	154	30	67	None	7.347225450204457e-10	9.133876633590017	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (30 total)
659	759	100	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
661	686	25	8	176	8	89	None	0.007267415029006458	2.1386200373214628	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019786-T01-IM6-74, ... (8 total)
668	718	50	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (13 total)
671	771	100	27	157	27	70	None	4.36030226790699e-09	8.36048340322365	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (27 total)
678	778	100	28	156	28	69	None	1.9355912689204797e-09	8.71318634557778	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (28 total)
681	781	100	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)
686	711	25	8	176	8	89	None	0.007267415029006458	2.1386200373214628	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019786-T01-IM6-74, ... (8 total)
686	736	50	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (13 total)
686	786	100	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)
686	886	200	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
693	718	25	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, EVT-P-0026407-T01-IM6-93, EVT-P-0045912-T02-IM6-182, EVT-P-0049318-T01-IM6-189
698	748	50	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (16 total)
702	752	50	17	167	17	80	None	1.0256347894034088e-05	4.98900725636106	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (17 total)
709	759	50	18	166	18	79	None	4.842124431071433e-06	5.314964054416519	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (18 total)
718	743	25	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, EVT-P-0026407-T01-IM6-93, EVT-P-0045912-T02-IM6-182, EVT-P-0049318-T01-IM6-189
718	768	50	18	166	18	79	None	4.842124431071433e-06	5.314964054416519	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (18 total)
718	818	100	21	163	21	76	None	4.937220617042471e-07	6.306517466132007	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (21 total)
718	918	200	21	163	21	76	None	4.937220617042471e-07	6.306517466132007	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (21 total)
721	771	50	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
723	748	25	11	173	11	86	None	0.0008432156599941764	3.0740613364461593	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (11 total)
727	752	25	12	172	12	85	None	0.00040846271224268893	3.388847583210815	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (12 total)
728	778	50	15	169	15	82	None	4.536095759828652e-05	4.343317785789269	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (15 total)
731	781	50	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (16 total)
734	759	25	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (13 total)
746	771	25	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (14 total)
748	773	25	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (14 total)
748	798	50	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (16 total)
748	848	100	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (16 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
748	948	200	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (16 total)
752	802	50	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0043363-T01-IM6-174, EVT-P-0045222-T03-IM7-179
752	852	100	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0043363-T01-IM6-174, EVT-P-0045222-T03-IM7-179
752	952	200	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0043363-T01-IM6-174, EVT-P-0045222-T03-IM7-179
756	781	25	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179
759	784	25	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179
759	809	50	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179
759	859	100	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179
759	959	200	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
23	223	200	87	97	1	96	0.00012112403100775194	7.679185037022372e-51	50.11468486756019	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, ... (87 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
113	313	200	72	112	1	96	0.002347417840375587	6.248379429212107e-34	33.20423260604981	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (72 total)
198	223	25	71	113	1	96	0.002529761904761905	4.096374544923265e-33	32.38760034125176	EVT-P-0000474-T01-IM3-9, EVT-P-0000557-T01-IM3-125, EVT-P-0000732-T01-IM3-65, EVT-P-0003206-T01-IM5-10, EVT-P-0003297-T01-IM5-11, ... (71 total)
486	686	200	76	108	75	22	293.1818181818182	4.406093942417991e-30	29.35594624748511	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002408-T02-IM5-8, EVT-P-0002613-T01-IM3-1, ... (76 total)
456	656	200	68	116	67	30	192.06666666666666	3.411263875176258e-25	24.467084684843417	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (68 total)
497	547	50	67	117	66	31	183.09677419354838	1.2835177750757548e-24	23.89159811252907	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (67 total)
482	532	50	65	119	64	33	166.78787878787878	1.7093128523025282e-23	22.76717844190544	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (65 total)
476	501	25	64	120	63	34	159.35294117647058	6.065285979848325e-23	22.217148717214226	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (64 total)
472	497	25	63	121	62	35	152.34285714285716	2.115676160934519e-22	21.67455080746973	EVT-P-0001532-T01-IM3-2, EVT-P-0001811-T01-IM3-3, EVT-P-0002217-T02-IM3-5, EVT-P-0002613-T01-IM3-1, EVT-P-0003383-T01-IM5-13, ... (63 total)
581	781	200	30	154	30	67	None	7.347225450204457e-10	9.133876633590017	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
681	781	100	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)
578	778	200	29	155	29	68	None	8.528605440240532e-10	9.069121976935888	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (29 total)
678	778	100	28	156	28	69	None	1.9355912689204797e-09	8.71318634557778	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (28 total)
571	771	200	28	156	28	69	None	1.9355912689204797e-09	8.71318634557778	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (28 total)
671	771	100	27	157	27	70	None	4.36030226790699e-09	8.36048340322365	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (27 total)
559	759	200	27	157	27	70	None	4.36030226790699e-09	8.36048340322365	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (27 total)
652	752	100	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)
659	759	100	26	158	26	71	None	9.751849301384782e-09	8.010913018636876	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (26 total)
648	748	100	25	159	25	72	None	2.165834310081195e-08	7.664374770688462	EVT-P-0002024-T01-IM3-4, EVT-P-0002408-T02-IM5-8, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0005786-T01-IM5-129, ... (25 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
718	818	100	21	163	21	76	None	4.937220617042471e-07	6.306517466132007	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (21 total)
63	113	50	16	168	0	97	0.0	2.809398109405692e-06	5.551386714174274	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (16 total)
709	759	50	18	166	18	79	None	4.842124431071433e-06	5.314964054416519	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (18 total)
45	70	25	15	169	0	97	0.0	6.59428167346614e-06	5.180832505991869	EVT-P-0001766-T04-IM6-151, EVT-P-0002266-T01-IM3-6, EVT-P-0012913-T02-IM6-137, EVT-P-0015646-T03-IM6-142, EVT-P-0020584-T01-IM6-79, ... (15 total)
702	752	50	17	167	17	80	None	1.0256347894034088e-05	4.98900725636106	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (17 total)
518	718	200	17	167	17	80	None	1.0256347894034088e-05	4.98900725636106	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0007705-T01-IM5-25, EVT-P-0010978-T01-IM5-36, ... (17 total)
698	748	50	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, ... (16 total)
731	781	50	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (16 total)
547	747	200	16	168	16	81	None	2.161926453446647e-05	4.665159084390838	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (16 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
728	778	50	15	169	15	82	None	4.536095759828652e-05	4.343317785789269	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0022138-T01-IM6-86, ... (15 total)
746	771	25	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (14 total)
618	718	100	14	170	14	83	None	9.475817175105265e-05	4.023383327076425	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (14 total)
734	759	25	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (13 total)
668	718	50	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0010978-T01-IM5-36, EVT-P-0013453-T01-IM5-46, ... (13 total)
501	701	200	13	171	13	84	None	0.000197125096463694	3.705258081139821	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, ... (13 total)
727	752	25	12	172	12	85	None	0.00040846271224268893	3.388847583210815	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (12 total)
723	748	25	11	173	11	86	None	0.0008432156599941764	3.0740613364461593	EVT-P-0002024-T01-IM3-4, EVT-P-0004218-T01-IM5-16, EVT-P-0005743-T02-IM5-152, EVT-P-0015152-T02-IM6-53, EVT-P-0025095-T01-IM6-147, ... (11 total)
636	686	50	9	175	9	88	None	0.003556113151803106	2.4490244286855454	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019560-T01-IM6-148, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
661	686	25	8	176	8	89	None	0.007267415029006458	2.1386200373214628	EVT-P-0002408-T02-IM5-8, EVT-P-0005786-T01-IM5-129, EVT-P-0006313-T01-IM5-19, EVT-P-0013453-T01-IM5-46, EVT-P-0019786-T01-IM6-74, ... (8 total)
693	718	25	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0010978-T01-IM5-36, EVT-P-0013886-T01-IM5-225, EVT-P-0026407-T01-IM6-93, EVT-P-0045912-T02-IM6-182, EVT-P-0049318-T01-IM6-189
752	802	50	5	179	5	92	None	0.060941084468509925	1.21508982161812	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0043363-T01-IM6-174, EVT-P-0045222-T03-IM7-179
756	781	25	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0022138-T01-IM6-86, EVT-P-0026657-T01-IM6-95, EVT-P-0038772-T01-IM6-163, EVT-P-0045222-T03-IM7-179
501	551	50	4	180	4	93	None	0.1231291066313099	0.9096392715597116	EVT-P-0007125-T01-IM5-23, EVT-P-0007705-T01-IM5-25, EVT-P-0034450-T01-IM6-120, EVT-P-0046720-T01-IM6-186

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002613-T01-IM3-1	P-0002613-T01-IM3	P-0002613	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42528199	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0001532-T01-IM3-2	P-0001532-T01-IM3	P-0001532	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525673	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001811-T01-IM3-3	P-0001811-T01-IM3	P-0001811	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525322	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EML4_c.3172+122:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002024-T01-IM3-4	P-0002024-T01-IM3	P-0002024	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42552725	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) inversion: c.2242+31:EML4_c.3173-739:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002217-T02-IM3-5	P-0002217-T02-IM3	P-0002217	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523244	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002266-T01-IM3-6	P-0002266-T01-IM3	P-0002266	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42473849	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002266-T01-IM3-7	P-0002266-T01-IM3	P-0002266	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	three_prime	42473852	3	70	True	70	True	exact match	retained	1.0	False	HELP motif			ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889:ALK_c.208+1022:EML4inv | Protein fusion: in frame (ALK-EML4)	NA
EVT-P-0002408-T02-IM5-8	P-0002408-T02-IM5	P-0002408	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42544452	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fuses with ALK exons 20-29) : c.2057-115:EML4_c.3172+274:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000474-T01-IM3-9	P-000047 4-T01-IM3	P-000047 4	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42497097	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003206-T01-IM5-10	P-000320 6-T01-IM5	P-000320 6	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42492359	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: mid-exon (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003297-T01-IM5-11	P-0003297-T01-IM5	P-0003297	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42504374	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003323-T01-IM5-12	P-0003323-T01-IM5	P-0003323	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42492435	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003383-T01-IM5-13	P-0003383-T01-IM5	P-0003383	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525071	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003446-T01-IM5-14	P-0003446-T01-IM5	P-0003446	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523253	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003913-T01-IM3-15	P-0003913-T01-IM3	P-0003913	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523334	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0004218-T01-IM5-16	P-0004218-T01-IM5	P-0004218	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42552850	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20): c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005305-T01-IM5-17	P-0005305-T01-IM5	P-0005305	Non-Small Cell Lung Cancer	SARCL	ALK-EML4	ALK	in-frame	five_prime	42502233	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0005321-T01-IM5-18	P-0005321-T01-IM5	P-0005321	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42528243	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20): c.2068+156:EML4_c.3173-693:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006313-T01-IM5-19	P-0006313-T01-IM5	P-0006313	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42544357	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063)-ALK (NM_004304) Fusion (EML4 exons 1 to 18 fused with exons 20 to 29 of ALK) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006370-T01-IM5-20	P-0006370-T01-IM5	P-0006370	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42526234	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006950-T01-IM5-21	P-0006950-T01-IM5	P-0006950	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42524958	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006950-T02-IM6-22	P-0006950-T02-IM6	P-0006950	Non-Small Cell Lung Cancer	LUNE	ALK-EML4	ALK	in-frame	five_prime	42524958	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29) : c.1489+2302:EML4_c.3172+943:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007125-T01-IM5-23	P-0007125-T01-IM5	P-0007125	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42528393	14	501	False	500	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328:EML4_c.3173-203:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0007211-T01-IM5-24	P-0007211-T01-IM5	P-0007211	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42492036	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165:EML4_c.3172+838:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007705-T01-IM5-25	P-0007705-T01-IM5	P-0007705	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42528485	14	532	False	531	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1594:EML4_c.3172+910:ALKinv | Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0008332-T01-IM5-26	P-0008332-T01-IM5	P-0008332	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	three_prime	42523484	14	497	True	497	True	exact match	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) rearrangement : c.1489+828:EML4_c.3173-518:ALKdup | Antisense fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009102-T01-IM5-27	P-0009102-T01-IM5	P-0009102	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42528137	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.1490-244:EML4_c.3173-119:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009301-T01-IM5-28	P-0009301-T01-IM5	P-0009301	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501297	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.668-6693:EML4_c.3173-257:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009301-T02-IM6-29	P-0009301-T02-IM6	P-0009301	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501297	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6693:EML4_c.3173-257:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009339-T01-IM5-30	P-0009339-T01-IM5	P-0009339	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501120	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009339-T02-IM5-31	P-0009339-T02-IM5	P-0009339	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501120	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-6870:EML4_c.3172+438:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009343-T01-IM5-32	P-0009343-T01-IM5	P-0009343	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42506897	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.668-1093:EML4_c.3173-127:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009353-T01-IM5-33	P-0009353-T01-IM5	P-0009353	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42494278	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.667+2407:EML4_c.3173-71:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0010041-T01-IM5-34	P-0010041-T01-IM5	P-0010041	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501013	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29) : c.701-6977:EML4_c.3172+385:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010177-T01-IM5-35	P-0010177-T01-IM5	P-0010177	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525732	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0010978-T01-IM5-36	P-0010978-T01-IM5	P-0010978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	out-of-frame	five_prime	42546901	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010978-T02-IM5-37	P-0010978-T02-IM5	P-0010978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	out-of-frame	five_prime	42546901	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv | Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0011018-T01-IM5-38	P-0011018-T01-IM5	P-0011018	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42506102	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011203-T01-IM5-39	P-0011203-T01-IM5	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	ALK	in-frame	five_prime	42501193	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011203-T02-IM5-40	P-0011203-T02-IM5	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	ALK	in-frame	five_prime	42501185	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011203-T03-IM6-41	P-0011203-T03-IM6	P-0011203	Salivary Gland Cancer	SDCA	ALK-EML4	ALK	in-frame	five_prime	42501139	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012679-T01-IM5-42	P-0012679-T01-IM5	P-0012679	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525468	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012778-T01-IM5-43	P-0012778-T01-IM5	P-0012778	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42493191	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012778-T02-IM6-44	P-0012778-T02-IM6	P-0012778	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42493191	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013202-T01-IM5-45	P-0013202-T01-IM5	P-0013202	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42494436	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565:EML4_c.3173-784:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013453-T01-IM5-46	P-0013453-T01-IM5	P-0013453	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42543741	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013864-T01-IM5-47	P-0013864-T01-IM5	P-0013864	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523314	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013869-T01-IM5-48	P-0013869-T01-IM5	P-0013869	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42528194	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014270-T01-IM6-49	P-0014270-T01-IM6	P-0014270	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42494165	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014310-T01-IM6-50	P-0014310-T01-IM6	P-0014310	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42497999	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20- 29): c.667+6128:EML4_c.3172+783:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T01-IM6-51	P-0014385-T01-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42524546	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015110-T01-IM6-52	P-0015110-T01-IM6	P-0015110	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42506264	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-1726:EML4_c.3173-124:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015152-T02-IM6-53	P-0015152-T02-IM6	P-0015152	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42552821	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:EML4_c.3173-685:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015197-T01-IM6-54	P-0015197-T01-IM6	P-0015197	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42505882	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20): c.668-2108:EML4_c.3172+318:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015434-T02-IM6-55	P-0015434-T02-IM6	P-0015434	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42522719	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015921-T03-IM6-56	P-0015921-T03-IM6	P-0015921	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42492387	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.667+516:EML4_c.3173-415:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015961-T01-IM6-57	P-0015961-T01-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501324	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015961-T02-IM6-58	P-0015961-T02-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501324	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015961-T03-IM6-59	P-0015961-T03-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAS	ALK-EML4	ALK	in-frame	five_prime	42501324	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016280-T01-IM6-60	P-0016280-T01-IM6	P-0016280	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525649	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016580-T01-IM6-61	P-0016580-T01-IM6	P-0016580	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42524913	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016960-T01-IM6-62	P-0016960-T01-IM6	P-0016960	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42499324	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017592-T01-IM5-63	P-0017592-T01-IM5	P-0017592	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42502027	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017978-T01-IM6-64	P-0017978-T01-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42499177	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000732-T01-IM3-65	P-0000732-T01-IM3	P-0000732	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42498669	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0017978-T03-IM6-66	P-0017978-T03-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42499177	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018242-T01-IM6-67	P-0018242-T01-IM6	P-0018242	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42500430	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused in-frame with ALK exons 20-29): c.668-7560:EML4_c.3172+320:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018280-T01-IM6-68	P-0018280-T01-IM6	P-0018280	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42501198	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 6 fused with ALK exon 20): c.494-6792:EML4_c.3173-793:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018969-T01-IM6-70	P-0018969-T01-IM6	P-0018969	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42527621	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-760:EML4_c.3083:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019491-T01-IM6-71	P-0019491-T01-IM6	P-0019491	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42527895	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490-486:EML4_c.3173-202:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019513-T01-IM6-72	P-0019513-T01-IM6	P-0019513	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42524934	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1489+2278:EML4_c.3096:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019533-T01-IM6-73	P-0019533-T01-IM6	P-0019533	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42498708	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837:EML4_c.3173-100:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019786-T01-IM6-74	P-0019786-T01-IM6	P-0019786	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42543272	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019972-T01-IM6-75	P-0019972-T01-IM6	P-0019972	Cancer of Unknown Primary	PDC	ALK-EML4	ALK	in-frame	five_prime	42493050	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020124-T01-IM6-76	P-0020124-T01-IM6	P-0020124	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523535	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020138-T01-IM6-77	P-0020138-T01-IM6	P-0020138	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42504974	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-3016:EML4_c.3172+420:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020408-T01-IM6-78	P-0020408-T01-IM6	P-0020408	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523883	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:EML4_c.3173-385:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020584-T01-IM6-79	P-0020584-T01-IM6	P-0020584	Renal Cell Carcinoma	URCC	ALK-EML4	ALK	in-frame	five_prime	42478795	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29) : c.209-4846:EML4_c.3173-654:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020780-T01-IM6-80	P-0020780-T01-IM6	P-0020780	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523934	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29): c.1489+1278:EML4_c.3173-690:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020825-T03-IM6-81	P-0020825-T03-IM6	P-0020825	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42477717	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29):c.208+4890:EML4_c.3173-373:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021257-T01-IM6-82	P-0021257-T01-IM6	P-0021257	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42543971	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1883-596:EML4_c.3172+550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0021495-T01-IM6-83	P-0021495-T01-IM6	P-0021495	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42494226	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021693-T01-IM6-84	P-0021693-T01-IM6	P-0021693	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42526872	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1316-1509:EML4_c.3173-608:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022101-T01-IM6-85	P-0022101-T01-IM6	P-0022101	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42544057	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510:EML4_c.2057-510:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022138-T01-IM6-86	P-0022138-T01-IM6	P-0022138	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42553384	21	778	False	777	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2333:EML4_c.3173-190:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0022398-T01-IM6-87	P-0022398-T01-IM6	P-0022398	Non-Small Cell Lung Cancer	NSCLCPD	ALK-EML4	ALK	in-frame	five_prime	42500137	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-7853:EML4_c.3172+292:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022926-T01-IM6-88	P-0022926-T01-IM6	P-0022926	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42492854	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+983:EML4_c.3173-164:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0023368-T01-IM6-89	P-0023368-T01-IM6	P-0023368	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42498875	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain): c.667+7004:EML4_c.3173-319:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023962-T01-IM6-90	P-0023962-T01-IM6	P-0023962	Cancer of Unknown Primary	ADNOS	ALK-EML4	ALK	in-frame	five_prime	42496646	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:EML4_c.3068-389:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024136-T01-IM6-91	P-0024136-T01-IM6	P-0024136	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42503492	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-4498:EML4_c.3172+61:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024907-T01-IM6-92	P-0024907-T01-IM6	P-0024907	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42482651	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29): c.209-990:EML4_c.3173-483:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0026407-T01-IM6-93	P-0026407-T01-IM6	P-0026407	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42550157	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1981-2450:EML4_c.3175:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026650-T02-IM6-94	P-0026650-T02-IM6	P-0026650	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	three_prime	42499245	7	223	True	223	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.667+7374:EML4_c.3172+876:ALKinv | Protein Fusion: in frame {ALK:EML4}	NA
EVT-P-0026657-T01-IM6-95	P-0026657-T01-IM6	P-0026657	Small Cell Lung Cancer	CSCLC	ALK-EML4	ALK	unknown	five_prime	42553363	21	771	False	770	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2312:EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026657-T03-IM6-96	P-0026657-T03-IM6	P-0026657	Small Cell Lung Cancer	SCLC	ALK-EML4	ALK	unknown	five_prime	42553363	21	771	False	770	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312:EML4_c.3173-558:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0027119-T01-IM6-97	P-0027119-T01-IM6	P-0027119	Thyroid Cancer	THPD	ALK-EML4	ALK	in-frame	five_prime	42527544	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027119-T03-IM6-98	P-0027119-T03-IM6	P-0027119	Thyroid Cancer	THPA	ALK-EML4	ALK	in-frame	five_prime	42527544	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027853-T01-IM6-99	P-0027853-T01-IM6	P-0027853	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42553137	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-157:EML4_c.3172+593:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028287-T01-IM6-100	P-0028287-T01-IM6	P-0028287	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42505367	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-2623:EML4_c.3173-267:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028339-T01-IM6-101	P-0028339-T01-IM6	P-0028339	Salivary Gland Cancer	OSACA	ALK-EML4	ALK	in-frame	five_prime	42474072	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+1245:EML4_c.3172+647:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028541-T01-IM6-102	P-0028541-T01-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42477458	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028541-T02-IM6-103	P-0028541-T02-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42477458	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028541-T03-IM6-104	P-0028541-T03-IM6	P-0028541	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42477458	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173-528:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028745-T01-IM6-105	P-0028745-T01-IM6	P-0028745	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525818	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1523-2563:EML4_c.3172:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028901-T01-IM6-106	P-0028901-T01-IM6	P-0028901	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42527537	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1316-844:EML4_c.3173-213:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029174-T01-IM6-107	P-0029174-T01-IM6	P-0029174	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42552862	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-19 fused with ALK exons 20-29): c.2242+168:EML4_c.3173-262:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029400-T01-IM6-108	P-0029400-T01-IM6	P-0029400	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523297	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+641:EML4_c.3172+659:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029550-T01-IM6-109	P-0029550-T01-IM6	P-0029550	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42507269	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029750-T01-IM6-110	P-0029750-T01-IM6	P-0029750	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42525807	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029971-T01-IM6-111	P-0029971-T01-IM6	P-0029971	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42507871	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-119:EML4_c.3172+520:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030590-T01-IM6-112	P-0030590-T01-IM6	P-0030590	Cancer of Unknown Primary	CUP	ALK-EML4	ALK	in-frame	five_prime	42482480	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKinv. | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031406-T01-IM6-113	P-0031406-T01-IM6	P-0031406	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42505101	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29): c.668-2889:EML4_c.3173-31:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031827-T01-IM6-114	P-0031827-T01-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42527708	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-673:EML4_c.3173-306:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T02-IM6-115	P-0031827-T02-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42527742	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032170-T01-IM6-116	P-0032170-T01-IM6	P-0032170	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42524478	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1822:EML4_c.3172+632:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032518-T01-IM6-117	P-0032518-T01-IM6	P-0032518	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42525263	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.3140:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032683-T01-IM6-118	P-0032683-T01-IM6	P-0032683	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42526259	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032862-T01-IM6-119	P-0032862-T01-IM6	P-0032862	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42523327	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion(EML4 exon 11 fused to ALK exon20): c.1489+671:EML4_c.3172+839:ALK | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034450-T01-IM6-120	P-0034450-T01-IM6	P-0034450	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	unknown	five_prime	42529087	14	547	True	547	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0034569-T02-IM6-121	P-0034569-T02-IM6	P-0034569	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42553077	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035121-T01-IM6-122	P-0035121-T01-IM6	P-0035121	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42503133	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4857:EML4_c.3172+944:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T01-IM6-123	P-0036106-T01-IM6	P-0036106	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42526326	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063)-ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036247-T01-IM6-124	P-0036247-T01-IM6	P-0036247	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42522763	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+107:EML4_c.3172+470:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000557-T01-IM3-125	P-0000557-T01-IM3	P-0000557	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42494981	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	NA | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017978-T02-IM6-126	P-0017978-T02-IM6	P-0017978	Non-Small Cell Lung Cancer	LUAD	ALK-EML4	ALK	in-frame	five_prime	42499177	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7306:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044042-T01-IM6-127	P-0044042-T01-IM6	P-0044042	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42506309	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1681:EML4_c.3172+600:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017592-T04-IM6-128	P-0017592-T04-IM6	P-0017592	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42502027	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-5963:EML4_c.3172+779:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005786-T01-IM5-129	P-0005786-T01-IM5	P-0005786	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42544381	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 with ALK exons 20-29) : c.2057-186:EML4_c.3172+88:ALKinv | Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T02-IM6-130	P-0014385-T02-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524546	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014385-T03-IM6-131	P-0014385-T03-IM6	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524546	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014385-T06-IM7-132	P-0014385-T06-IM7	P-0014385	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524546	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1890:EML4_c.3172+651:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022184-T01-IM6-133	P-0022184-T01-IM6	P-0022184	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524042	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1315+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022184-T04-IM6-134	P-0022184-T04-IM6	P-0022184	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524042	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1386:EML4_c.3173-641:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012873-T02-IM6-135	P-0012873-T02-IM6	P-0012873	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524887	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012873-T03-IM7-136	P-0012873-T03-IM7	P-0012873	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524887	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012913-T02-IM6-137	P-0012913-T02-IM6	P-0012913	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42479508	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-4133:EML4_c.3173-571:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023375-T02-IM7-138	P-0023375-T02-IM7	P-0023375	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42525653	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2728:EML4_c.3172+473:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031368-T04-IM6-139	P-0031368-T04-IM6	P-0031368	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527227	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154:EML4_c.3173-797:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008332-T02-IM6-140	P-0008332-T02-IM6	P-0008332	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42523484	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173-519:ALKdup | Antisense Fusion	NA
EVT-P-0031824-T01-IM6-141	P-0031824-T01-IM6	P-0031824	Pancreatic Cancer	PAAD	EML4-ALK	ALK	in-frame	five_prime	42475798	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971:EML4_c.2816-19:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015646-T03-IM6-142	P-0015646-T03-IM6	P-0015646	Nerve Sheath Tumor	MPNST	EML4-ALK	ALK	in-frame	five_prime	42479732	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.209-3909:EML4_c.3172+299:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0013299-T02-IM6-143	P-0013299-T02-IM6	P-0013299	Breast Cancer	IDC	EML4-ALK	ALK	in-frame	five_prime	42493173	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063)-ALK (NM_004304) Fusion(EML4 exon5 fused with ALK exon20) : c.667+1302:EML4_c.3173-644:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031827-T03-IM6-144	P-0031827-T03-IM6	P-0031827	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527742	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031827-T06-IM7-145	P-0031827-T06-IM7	P-0031827	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527742	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-639:EML4_c.3173-252:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025040-T01-IM6-146	P-0025040-T01-IM6	P-0025040	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526441	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20): c.1316-1940:EML4_c.3173-758:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0025095-T01-IM6-147	P-0025095-T01-IM6	P-0025095	Non-Small Cell Lung Cancer	NSCLC	EML4-ALK	ALK	in-frame	five_prime	42553016	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-278:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019560-T01-IM6-148	P-0019560-T01-IM6	P-0019560	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	out-of-frame	five_prime	42540223	17	656	True	655	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968-2879:EML4_c.3172+375:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0032683-T02-IM7-149	P-0032683-T02-IM7	P-0032683	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526259	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2122:EML4_c.3172+564:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034334-T01-IM6-150	P-0034334-T01-IM6	P-0034334	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496780	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4909:EML4_c.3173-176:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001766-T04-IM6-151	P-0001766-T04-IM6	P-0001766	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42476789	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3962:EML4_c.3068-336:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005743-T02-IM5-152	P-0005743-T02-IM5	P-0005743	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42553113	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243-181:EML4_c.3172+658:ALKinv | Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0035085-T03-IM6-153	P-0035085-T03-IM6	P-0035085	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42499232	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7361:EML4_c.3173-685:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015961-T07-IM6-154	P-0015961-T07-IM6	P-0015961	Non-Small Cell Lung Cancer	LUAS	EML4-ALK	ALK	in-frame	five_prime	42501324	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6666:EML4_c.3173-132:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036106-T03-IM7-155	P-0036106-T03-IM7	P-0036106	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526326	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-536:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027119-T04-IM7-156	P-0027119-T04-IM7	P-0027119	Thyroid Cancer	THPD	EML4-ALK	ALK	in-frame	five_prime	42527544	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-837:EML4_c.3173-317:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037404-T01-IM6-157	P-0037404-T01-IM6	P-0037404	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496834	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4963:EML4_c.3172+455:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037491-T01-IM6-158	P-0037491-T01-IM6	P-0037491	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527261	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon12 fused with ALK exon20): c.1490-1120:EML4_c.3173-392:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037668-T01-IM6-159	P-0037668-T01-IM6	P-0037668	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42525639	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2742:EML4_c.3172+696:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037925-T01-IM6-160	P-0037925-T01-IM6	P-0037925	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496505	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4634:EML4_c.3173-128:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037982-T01-IM6-161	P-0037982-T01-IM6	P-0037982	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42528243	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-138:EML4_c.3173-85:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038201-T03-IM6-162	P-0038201-T03-IM6	P-0038201	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42523645	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+989:EML4_c.3172+579:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038772-T01-IM6-163	P-0038772-T01-IM6	P-0038772	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42553539	21	781	True	780	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2341+147:EML4_c.3173-293:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040261-T01-IM6-164	P-0040261-T01-IM6	P-0040261	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527041	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1340:EML4_c.3173-709:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040553-T01-IM6-165	P-0040553-T01-IM6	P-0040553	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42522686	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+30:EML4_c.3173-661:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040574-T01-IM6-166	P-0040574-T01-IM6	P-0040574	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42506401	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589:EML4_c.3172+589:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040576-T01-IM6-167	P-0040576-T01-IM6	P-0040576	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42507169	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:EML4_c.3173-927:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040600-T01-IM6-168	P-0040600-T01-IM6	P-0040600	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42493666	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1795:EML4_c.3172+639:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040732-T02-IM6-169	P-0040732-T02-IM6	P-0040732	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526549	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1832:EML4_c.3173-242:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041967-T01-IM6-170	P-0041967-T01-IM6	P-0041967	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42523103	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041967-T02-IM7-171	P-0041967-T02-IM7	P-0041967	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42523103	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+447:EML4_c.3172+180:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043041-T01-IM6-172	P-0043041-T01-IM6	P-0043041	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524438	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1782:EML4_c.3173-203:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043350-T01-IM6-173	P-0043350-T01-IM6	P-0043350	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42503973	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4017:EML4_c.3172+814:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043363-T01-IM6-174	P-0043363-T01-IM6	P-0043363	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42553305	21	752	False	751	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2254:EML4_c.3172+226:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0043594-T01-IM6-175	P-0043594-T01-IM6	P-0043594	Colorectal Cancer	MACR	EML4-ALK	ALK	unknown	five_prime	42486142	3	113	True	112	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.339-2119:EML4_c.3118:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007705-T02-IM6-176	P-0007705-T02-IM6	P-0007705	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42528485	14	532	False	531	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1594:EML4_c.3172+910:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0061297-T03-IM7-177	P-0061297-T03-IM7	P-0061297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42523587	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014310-T02-IM6-178	P-0014310-T02-IM6	P-0014310	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42497999	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+6128:EML4_c.3172+783:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045222-T03-IM7-179	P-0045222-T03-IM7	P-0045222	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42553328	21	759	False	759	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045243-T01-IM6-180	P-0045243-T01-IM6	P-0045243	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42504926	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045341-T01-IM6-181	P-0045341-T01-IM6	P-0045341	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42503556	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045912-T02-IM6-182	P-0045912-T02-IM6	P-0045912	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	out-of-frame	five_prime	42545675	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011:EML4_c.3173-113:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0046083-T01-IM6-183	P-0046083-T01-IM6	P-0046083	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42523642	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046083-T02-IM7-184	P-0046083-T02-IM7	P-0046083	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42523642	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+986:EML4_c.3173-271:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046594-T01-IM6-185	P-0046594-T01-IM6	P-0046594	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42494088	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2217:EML4_c.3173-51:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046720-T01-IM6-186	P-0046720-T01-IM6	P-0046720	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	out-of-frame	five_prime	42528533	14	547	True	547	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1641+1:EML4_c.3173-275:ALKinv | Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0047355-T01-IM6-187	P-0047355-T01-IM6	P-0047355	Non-Small Cell Lung Cancer	LUSC	EML4-ALK	ALK	in-frame	five_prime	42492796	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047355-T02-IM6-188	P-0047355-T02-IM6	P-0047355	Non-Small Cell Lung Cancer	LUSC	EML4-ALK	ALK	in-frame	five_prime	42492796	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+925:EML4_c.3173-215:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049318-T01-IM6-189	P-0049318-T01-IM6	P-0049318	Thyroid Cancer	THHC	EML4-ALK	ALK	unknown	five_prime	42546999	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+2335:EML4_c.3208:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049375-T01-IM6-190	P-0049375-T01-IM6	P-0049375	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42493037	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049375-T02-IM6-191	P-0049375-T02-IM6	P-0049375	Non-Small Cell Lung Cancer	SARCL	EML4-ALK	ALK	in-frame	five_prime	42493037	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1166:EML4_c.3173-338:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049511-T01-IM6-192	P-0049511-T01-IM6	P-0049511	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42552703	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+9:EML4_c.3173-597:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051231-T01-IM6-193	P-0051231-T01-IM6	P-0051231	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496001	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051231-T03-IM6-194	P-0051231-T03-IM6	P-0051231	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496001	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4130:EML4_c.3173-380:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0052035-T03-IM6-195	P-0052035-T03-IM6	P-0052035	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42552773	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2242+79:EML4_c.3172+671:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052297-T02-IM6-196	P-0052297-T02-IM6	P-0052297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42528220	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-161:EML4_c.3172+72:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054263-T01-IM6-197	P-0054263-T01-IM6	P-0054263	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42507780	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-210:EML4_c.3173-163:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054408-T01-IM6-198	P-0054408-T01-IM6	P-0054408	Non-Small Cell Lung Cancer	LUAS	EML4-ALK	ALK	in-frame	five_prime	42553032	20	748	True	747	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2243-262:EML4_c.3173-508:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0057391-T02-IM6-199	P-0057391-T02-IM6	P-0057391	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42504972	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3018:EML4_c.3173-698:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058599-T02-IM7-200	P-0058599-T02-IM7	P-0058599	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527036	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1345:EML4_c.3173-21:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059257-T02-IM7-201	P-0059257-T02-IM7	P-0059257	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42507008	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-982:EML4_c.3172+117:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059699-T03-IM7-202	P-0059699-T03-IM7	P-0059699	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526391	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1990:EML4_c.3172+860:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0060148-T01-IM7-203	P-0060148-T01-IM7	P-0060148	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42525256	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2600:EML4_c.3172+570:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0061297-T02-IM7-204	P-0061297-T02-IM7	P-0061297	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42523587	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+931:EML4_c.3124:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0010177-T02-IM7-205	P-0010177-T02-IM7	P-0010177	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42525732	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2649:EML4_c.3173-964:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062985-T02-IM7-206	P-0062985-T02-IM7	P-0062985	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42496432	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+4561:EML4_c.3173-805:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062999-T02-IM7-207	P-0062999-T02-IM7	P-0062999	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42526326	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-2055:EML4_c.3173-697:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0063585-T01-IM7-208	P-0063585-T01-IM7	P-0063585	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42505177	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2813:EML4_c.3173-490:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064342-T01-IM7-209	P-0064342-T01-IM7	P-0064342	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42476869	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+4042:EML4_c.3173-140:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064716-T02-IM7-210	P-0064716-T02-IM7	P-0064716	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42504321	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669:EML4_c.3172+588:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065210-T01-IM7-211	P-0065210-T01-IM7	P-0065210	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42505978	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065469-T02-IM7-212	P-0065469-T02-IM7	P-0065469	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42506041	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0066568-T02-IM7-213	P-0066568-T02-IM7	P-0066568	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42495299	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+3428:EML4_c.3172+255:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067048-T02-IM7-214	P-0067048-T02-IM7	P-0067048	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42494087	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+2216:EML4_c.3172+936:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067069-T01-IM7-215	P-0067069-T01-IM7	P-0067069	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524961	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2305:EML4_c.3172+158:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067093-T02-IM7-216	P-0067093-T02-IM7	P-0067093	Renal Cell Carcinoma	URCC	EML4-ALK	ALK	in-frame	five_prime	42474335	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+1508:EML4_c.3173-909:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0068467-T02-IM7-217	P-0068467-T02-IM7	P-0068467	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42524222	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+1566:EML4_c.3172+32:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0069396-T02-IM7-218	P-0069396-T02-IM7	P-0069396	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42503682	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4308:EML4_c.3173-550:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069535-T01-IM7-219	P-0069535-T01-IM7	P-0069535	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42476656	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+3829:EML4_c.3172+563:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

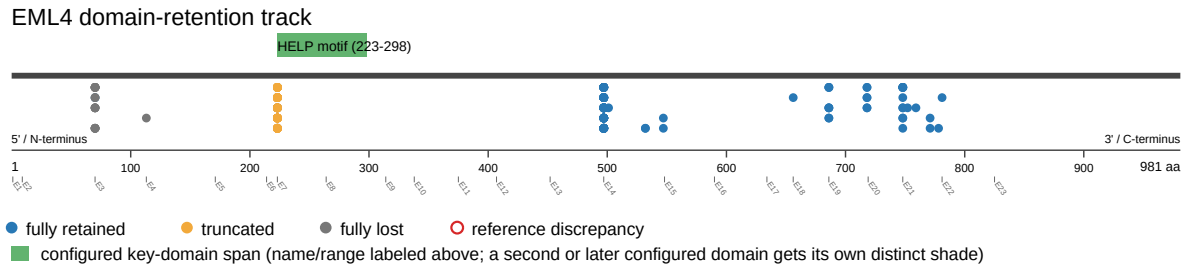
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0069617-T01-IM7-220	P-0069617-T01-IM7	P-0069617	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42499692	6	223	True	222	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.013157894736842105	True			HELP motif	EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+7821:EML4_c.3172+310:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043860-T02-IM6-221	P-0043860-T02-IM6	P-0043860	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	in-frame	five_prime	42527401	13	497	True	496	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-980:EML4_c.3173-537:ALKinv | Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044432-T01-IM6-222	P-0044432-T01-IM6	P-0044432	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	ALK	unknown	five_prime	42543991	18	686	True	685	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	HELP motif			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2057-576:EML4_c.3155:ALKinv | Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0059139-T01-IM7-224	P-0059139-T01-IM7	P-0059139	Pancreatic Cancer	PAAD	EML4-NTRK1	NTRK1	in-frame	five_prime	42473987	2	70	True	69	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		HELP motif		EML4 (NM_019063) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;p21)(chr1:g.156844256::chr2:g.42473987) | Protein Fusion: in frame {EML4:NTRK1}	NA

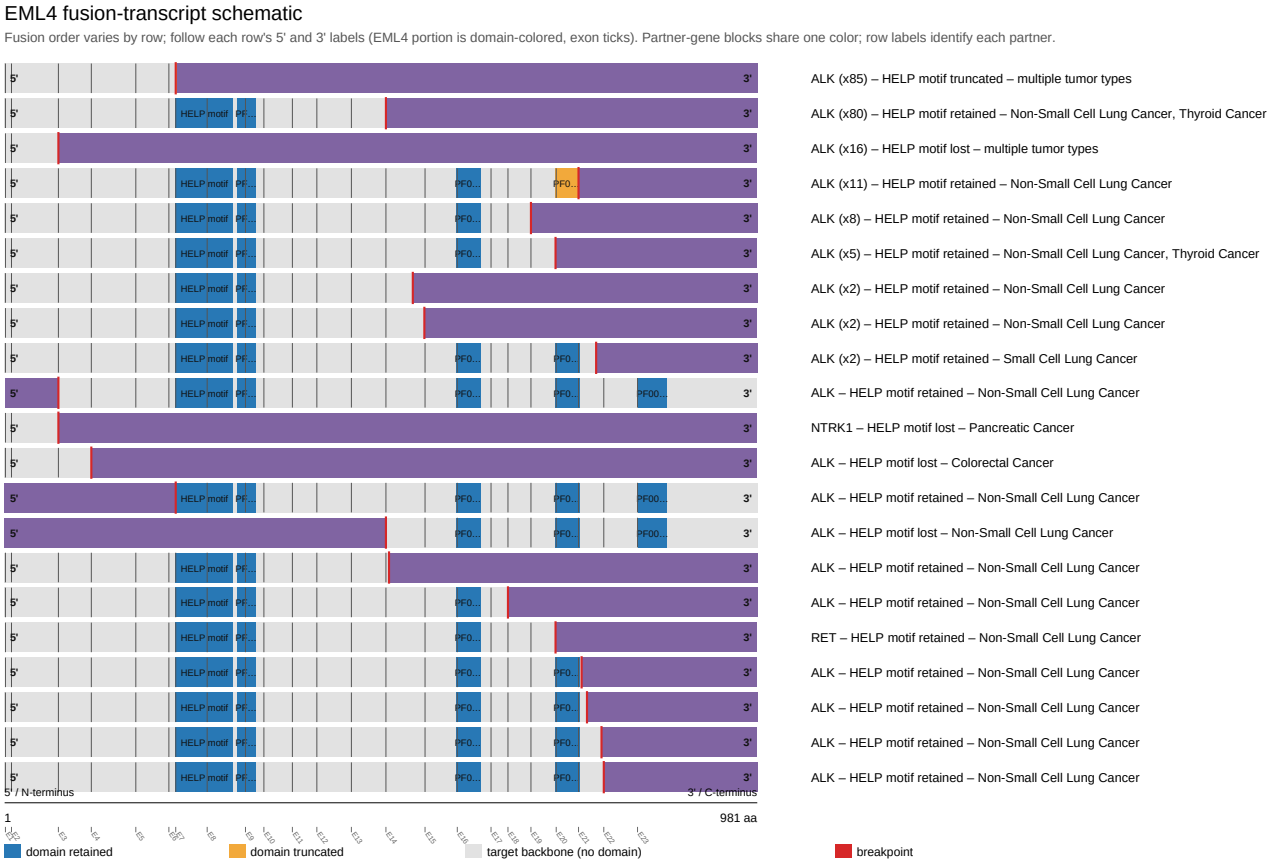
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013886-T01-IM5-225	P-0013886-T01-IM5	P-0013886	Non-Small Cell Lung Cancer	LUAD	RET-EML4	RET	unknown	five_prime	42550559	19	718	True	718	True	exact match	retained	1.0	False	HELP motif			EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) | Protein fusion: mid-exon (EML4-RET)	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.